



IMA205 Challenge 2025

Cardiac Pathology Prediction

Final Report

Author:
Mariana Olm Rezende

April 2025

Contents

1	Introduction	2
1.1	Motivation	2
1.2	Objective	2
2	Data	2
3	Left Ventricle Segmentation	4
3.1	Hypothesis: Filling the Myocardium to Reconstruct the LV	4
3.2	Segmentation Strategy and Results	5
4	Features	6
4.1	Basic Features	6
4.1.1	Body Surface Area (BSA)	6
4.1.2	Cardiac Chamber Volume	6
4.1.3	Myocardium Thickness	7
4.2	Feature Engineering	7
4.2.1	Normalized Volume	7
4.2.2	Ejection Fraction (EF)	8
4.2.3	Ratios	9
5	Classification Model: Random Forest	10
5.1	Data and Features	10
5.2	Modelling Method	11
6	Results	13
7	Conclusion	14

1 Introduction

1.1 Motivation

Cardiac function analysis is an essential component of clinical cardiology, playing a critical role in diagnosing heart diseases, managing patient care, and guiding informed treatment decisions. Early detection of specific cardiac conditions is vital, as delayed diagnosis can lead to severe health complications, including heart failure and sudden cardiac arrest. This challenge [2] addresses the importance of identifying four significant cardiac conditions that often remain undetected until advanced stages, emphasizing the necessity of timely and accurate diagnosis.

In recent years, Cardiac Magnetic Resonance Imaging (CMRI) has emerged as a highly effective, non-invasive diagnostic tool, allowing clinicians to detect and assess cardiac pathologies with more precision.

1.2 Objective

The primary objective of this challenge is to classify patients based on their CMRI exams into one of five diagnostic categories:

1. Healthy controls (NOR)
2. Myocardial infarction (MINF)
3. Dilated cardiomyopathy (DCM)
4. Hypertrophic cardiomyopathy (HCM)
5. Abnormal right ventricle (ARV)

Accurate classification allows for earlier and more effective treatment, improving patient outcomes.

2 Data

The dataset used in this challenge is derived from the ACDC dataset ¹, comprising cardiac MRI scans from 150 subjects. The data is split into a training set of 100 subjects and a testing set of 50. For the training data, the classes are evenly distributed, with 20 subjects for each of the five diagnostic categories: healthy controls (NOR), myocardial infarction (MINF), dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), and abnormal right ventricle (ARV).

Each subject includes volumetric MRI data stored in NIfTI format, which preserves 3D spatial information alongside header metadata such as voxel spacing, accessible through functions like `get_zooms()`. Imaging was performed at two key phases of the cardiac cycle: end diastole (ED)

¹O. Bernard, A. Lalande, C. Zotti, F. Cervenansky, et al., “Deep Learning Techniques for Automatic MRI Cardiac Multi-structures Segmentation and Diagnosis: Is the Problem Solved?”, *IEEE Transactions on Medical Imaging*, vol. 37, no. 11, pp. 2514–2525, Nov. 2018. doi: 10.1109/TMI.2018.2837502.

and end systole (ES), allowing dynamic assessment of cardiac function and structure. Additionally, patient metadata, specifically height and weight, was provided, enabling further analysis of anatomical and physiological variations across subjects.

In each NIfTI file, two types of information are available for every slice: the original MRI image data, capturing the anatomy of the heart, and the corresponding segmentation matrices that annotate key anatomical structures. These manual segmentation masks label four classes: background (0), right ventricle cavity (1), myocardium (2), and left ventricle cavity (3). This dual information forms the basis for both training segmentation models and subsequent clinical diagnosis tasks.

Figure 1 presents representative slices showing segmented structures across both ED and ES phases, highlighting the distinct regions used for the analysis.

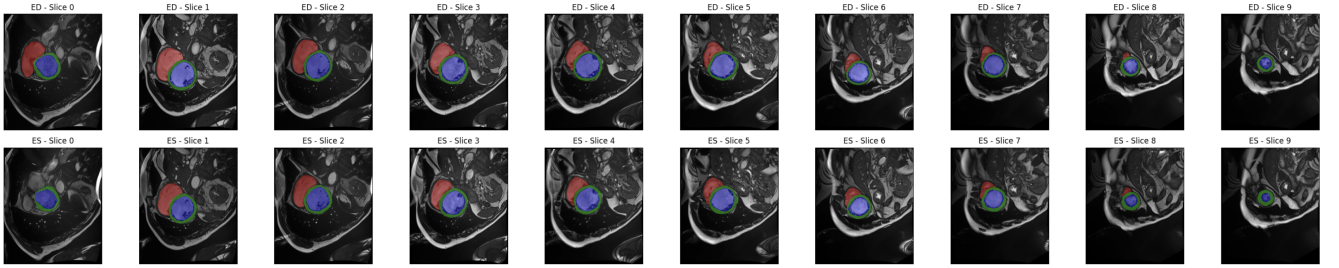


Figure 1: Example slices showing cardiac segmentations in ED and ES phases across multiple slices. Segmentations include: right ventricle cavity (red), myocardium (green), and left ventricle cavity (blue).

Before proceeding with model development, we first assessed the availability and completeness of the segmentation masks across the dataset. In particular, we verified how many subjects had complete annotations for each anatomical structure in both ED and ES phases, for both training and test splits. The results of this analysis are summarized in Table 1.

Table 1: Number of subjects with available segmentation labels per structure, phase, and data split.

Data Split	Phase	Background	Right ventricle cavity	Myocardium	Left ventricle cavity
Train	ED	100	100	100	100
	ES	100	100	100	100
Test	ED	50	50	50	0
	ES	50	50	50	0

As shown in Table 1, all four cardiac structures are fully segmented in the training set across both cardiac phases. However, in the test set, segmentation labels are only available for the background, right ventricle cavity, and myocardium. Notably, the left ventricle cavity (LV) is not segmented in any of the test volumes.

This lack of LV annotations presents a major challenge, particularly because left ventricular metrics are among the most important indicators in cardiac pathology classification. Without these values, it is not possible to compute several key features directly from the test data, limiting the performance of any diagnostic model.

To address this limitation, we implemented an additional step to generate left ventricle segmentations for the test set, enabling consistent metric extraction and classification across both training and test datasets.

3 Left Ventricle Segmentation

By examining examples from the training set, where full annotations are available, we observed that the myocardium (label 2) consistently forms a closed boundary around the LV cavity. This pattern suggested a potential approach: the region enclosed by the myocardium could approximate the location of the LV.

Figure 2 illustrates examples of test subject slices across ED and ES phases where the myocardium is clearly visible, but the LV is not labeled.

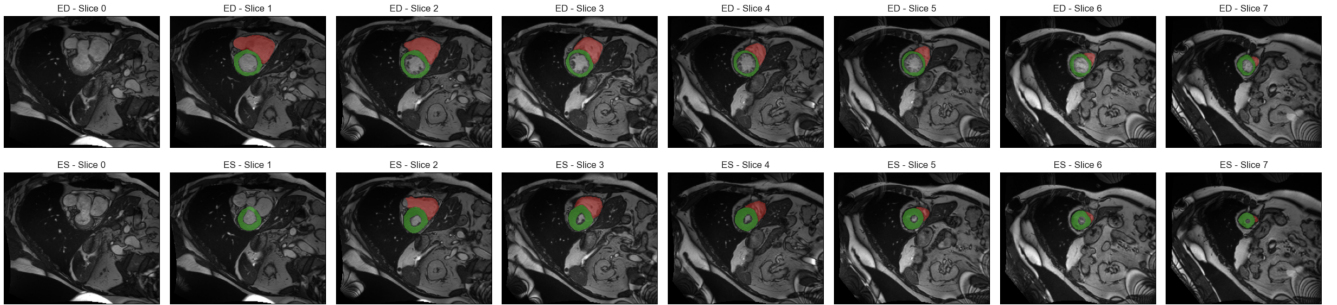


Figure 2: Example test subject slices across ED and ES phases with visible myocardium but missing left ventricle (LV) segmentation.

3.1 Hypothesis: Filling the Myocardium to Reconstruct the LV

Based on this observation, we hypothesized that the region enclosed by the myocardium corresponds to the LV cavity. To test this hypothesis, we applied a morphological operation to fill the enclosed areas within the myocardium on the training data, where ground truth LV labels are available.

Specifically, we used the `binary_fill_holes` function from `scipy.ndimage`, which identifies and fills all internal cavities (holes) within a binary object. In our case, it was applied to the myocardium mask (label 2), reconstructing the interior LV region and reassigning it with the LV label (label 3). The resulting modified segmentations were then compared against the original LV masks using the Intersection over Union (IoU) metric.

Initial results showed that this method accurately reconstructed the LV in most cases, achieving high IoU scores. However, a subset of slices failed to produce a valid LV region. All failure cases, shown in Figure 3, have gaps in the myocardial contour, meaning the region is not fully enclosed. Since `binary_fill_holes` depends on closed boundaries to identify fillable areas, it cannot handle these cases correctly.

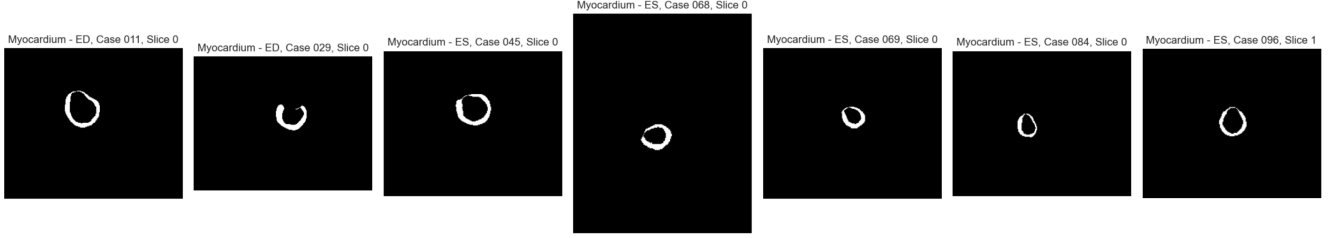


Figure 3: Examples of slices where the LV region could not be filled due to open myocardium contours.

To address this issue, we applied a convex hull operation to the myocardium mask. The convex hull is the smallest convex shape that fully contains the original object, effectively “closing” gaps or missing edges in the myocardium contour. By computing the convex hull of the myocardium, we generate a smooth, fully enclosed boundary.

Next, we subtract the original myocardium mask from its convex hull. This subtraction isolates the internal region corresponding to the potential LV cavity. Finally, to reduce noise and remove small spurious areas, we retain only the largest connected component within this difference mask.

3.2 Segmentation Strategy and Results

The convex hull solution was applied to all previously failed slices, yielding the following corrected masks:

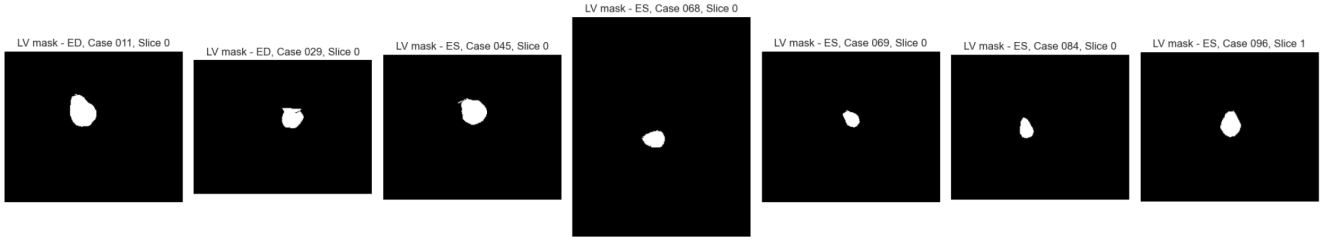


Figure 4: Filled LV regions using the convex hull method.

The result in Figure 4 shows that the proposed strategy successfully restores the left ventricle (LV) region. By using the convex hull, the method creates a closed contour, allowing the LV to be filled correctly even when the original myocardium mask is incomplete.

To quantify the effectiveness of the proposed segmentation strategy, the method was applied to the full training set. The evaluation was based on the Intersection over Union (IoU) metric, computed for each slice that contained both a myocardium and a corresponding ground truth LV label. This restriction ensured that the metric was only calculated where a valid comparison was possible. The IoU is defined as:

$$\text{IoU} = \frac{|\text{Prediction} \cap \text{Ground Truth}|}{|\text{Prediction} \cup \text{Ground Truth}|}$$

The results indicated strong performance of the segmentation method. Among all evaluated slices, only one case had an IoU score below the predefined threshold of 0.9, with a value of 0.800. All other slices achieved IoU scores above 0.9, resulting in an overall mean IoU of 0.9998 across the dataset. These results highlight the robustness and accuracy of the proposed approach in recovering the LV region using morphological operations, even in the presence of slight anatomical imperfections.

Given the strong performance on the training set, the same method was extended to the test dataset to generate the missing left ventricle masks. Figure 5 shows the result of this procedure applied to the same test subject previously illustrated in Figure 2.

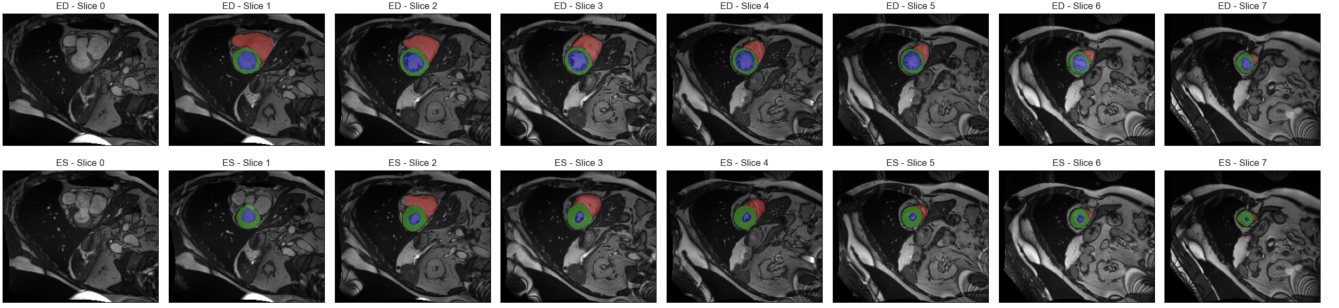


Figure 5: Test subject slices with reconstructed LV regions based on convex hull segmentation.

4 Features

To characterize cardiac morphology and function, a series of clinical and image derived features were computed for each subject. These features serve as inputs to the classification model and are consistent with established cardiology literature, particularly the diagnostic criteria described by Bernard et al.[1] and Wolterink et al.[4].

4.1 Basic Features

4.1.1 Body Surface Area (BSA)

One of the foundational subject-specific metrics is the body surface area (BSA), which is used to normalize volumetric measurements and account for inter-patient variability in body size. BSA was computed for each subject using the Du Bois formula:

$$BSA = 0.007184 \times \text{Weight}^{0.425} \times \text{Height}^{0.725}$$

where weight is expressed in kilograms and height in centimeters.

4.1.2 Cardiac Chamber Volume

Cardiac chamber volumes were calculated by multiplying the number of voxels in each segmented region by the physical volume of a single voxel. The voxel volume was derived from the NIfTI metadata using voxel spacing values. Formally, the volume V of a region is defined as:

$$V = N_{\text{voxels}} \times V_{\text{voxel}}$$

Volumes were computed separately for the left ventricle (LV), right ventricle (RV), and myocardium (MYO) at both the end-diastole (ED) and end-systole (ES) phases.

4.1.3 Myocardium Thickness

Myocardium thickness was computed to quantify structural remodeling in pathological conditions such as hypertrophic cardiomyopathy. Thickness was measured at end-diastole (ED), the phase where myocardial boundaries are best defined and most clinically relevant. For each subject, the mean and maximum thickness were calculated by extracting inner and outer myocardial contours from the segmentation masks and computing pairwise distances in physical space using voxel spacing information. This was performed for each slice using a combination of OpenCV contour detection and Euclidean distance matrices.

The algorithm first identifies the inner and outer myocardial contours and converts their coordinates from pixel to real-world units (millimeters). Then, the minimum distance from each point on the inner contour to the outer contour is calculated. From these distances, both the maximum and average myocardial thicknesses per subject are extracted. These metrics are later used in classification and correlated with known patterns of myocardial hypertrophy or thinning associated with specific cardiac pathologies.

4.2 Feature Engineering

Several new features were derived from the basic features to better reflect clinical patterns linked to different heart conditions. These included normalized values and ratios, making it easier to compare subjects with varying body sizes and heart shapes. The following subsections describe the main engineered features used in the classification model.

4.2.1 Normalized Volume

To account for differences in body size among subjects, end-diastolic (ED) volumes of the left and right ventricles were normalized by each individual’s body surface area (BSA), resulting in a dimensionally consistent metric:

$$\text{Normalized Volume} = \frac{V_{\text{ED}}}{\text{BSA}}$$

Figure 6 illustrates the distribution of normalized ED volumes across different diagnostic categories. The boxplot results align with the diagnostic ranges presented by Bernard et al. [1]. In the NOR group, LV diastolic volumes remain below 90 mL/m² and RV volumes are under 100 mL/m². The MINF group exhibits increased LV volumes, characteristic of post-infarction remodeling. In the DCM group, LV volumes exceed 100 mL/m², reflecting significant ventricular dilation. The ARV group shows markedly elevated RV volumes, often exceeding 100 mL/m², while maintaining relatively normal LV volumes.

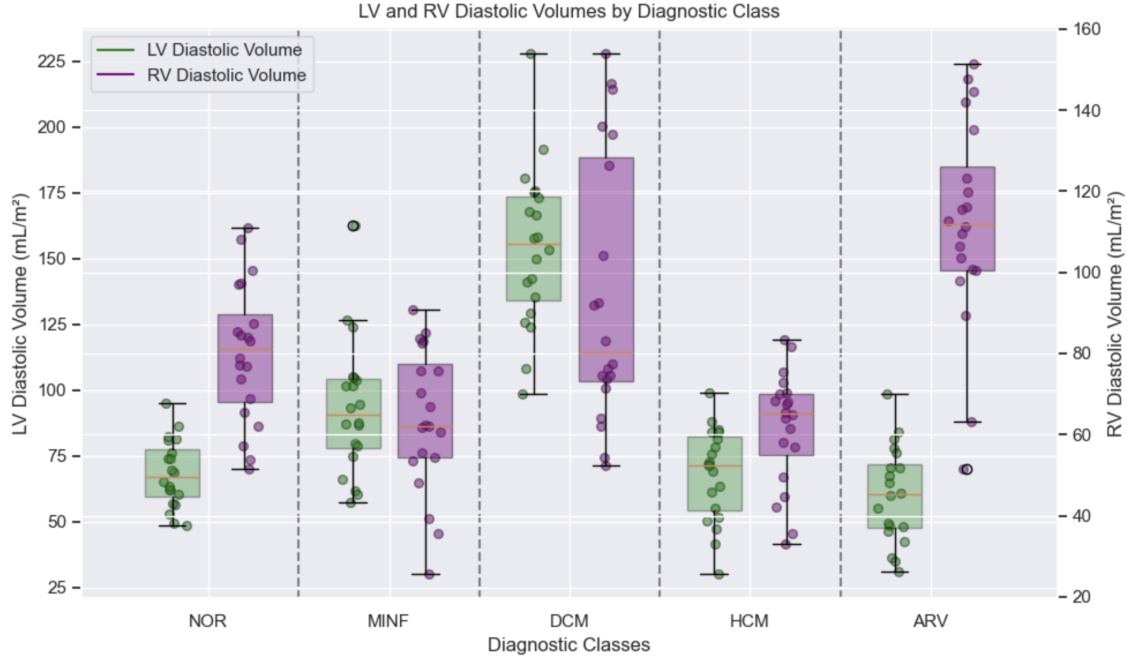


Figure 6: Boxplots of normalized LV and RV end-diastolic volumes (in mL/m²) across diagnostic categories.

4.2.2 Ejection Fraction (EF)

Another key measure is the ejection fraction (EF), which represents the heart's pumping efficiency. It is defined as the proportion of blood ejected from a ventricle during a single cardiac cycle:

$$EF = \frac{EDV - ESV}{EDV}$$

where EDV and ESV are the end-diastolic and end-systolic volumes, respectively.

This feature was calculated for both the left and right ventricles, giving LVEF and RVEF. Figure 7 shows how these values vary across the diagnostic groups. In the NOR group, LVEF is above 50% and RVEF above 40%, which is considered normal. MINF and DCM subjects show much lower LVEFs, often under 40%, which points to impaired systolic function. In contrast, the HCM group has LVEFs above 55%, which is expected in hypertrophic cardiomyopathy. ARV subjects have normal LVEFs but their RVEFs are often below 40%, suggesting isolated dysfunction of the right ventricle.

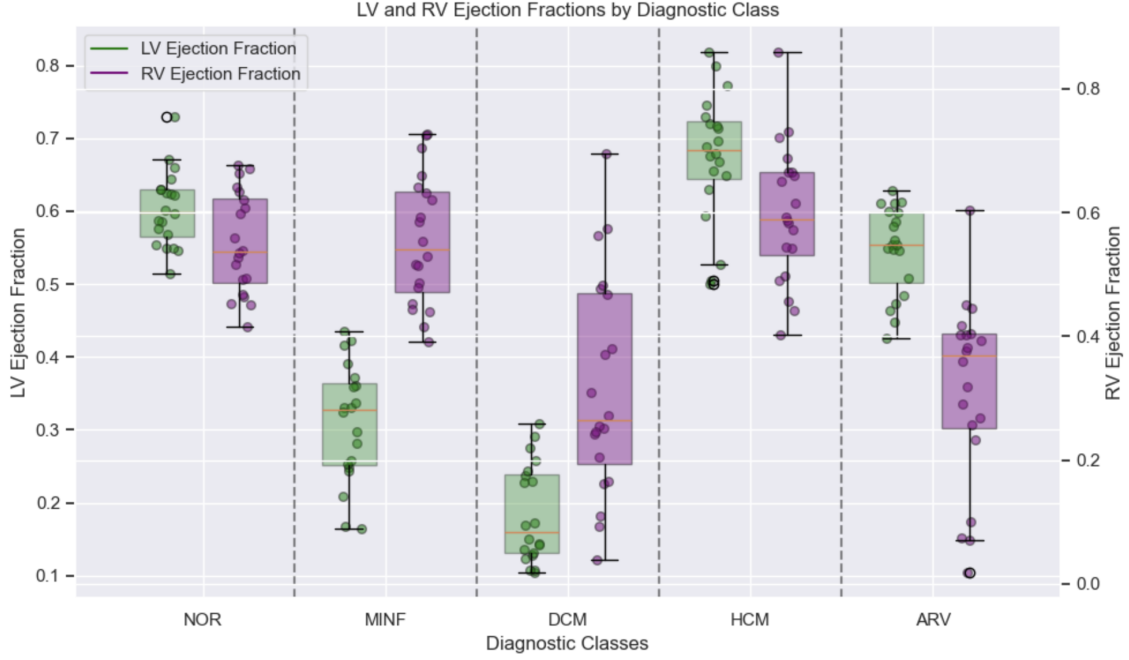


Figure 7: Boxplots of left and right ventricular ejection fractions (LVEF, RVEF) across diagnostic categories.

4.2.3 Ratios

In addition to absolute volumetric features, relative measures were computed to capture the relationship between different cardiac structures. Following the approach described in [4], the following ratio features were derived:

- $V[RV]/V[LV]$ at ED
- $V[RV]/V[LV]$ at ES
- $V[MYO]/V[LV]$ at ED
- $V[MYO]/V[LV]$ at ES

These ratios serve several purposes. First, the RV/LV volume ratios at both ED and ES are indicators of relative chamber size. Changes in these ratios can be indicative of right ventricular enlargement or left ventricular underperformance, which are relevant in conditions where ventricular balance is disrupted. Second, the MYO/LV volume ratios provide insight into the relative thickness of the myocardium in relation to the ventricular cavity. This is especially important in the context of hypertrophic cardiomyopathy, where an abnormally thick myocardium contrasts with a smaller ventricular volume.

Including these ratio features in the classification model helps capture not just the size of each structure, but also how they relate to each other. These relationships can sometimes reflect pathological changes more clearly than absolute values alone. In practice, they added useful information about the heart's overall geometry and function, as also shown in [4].

5 Classification Model: Random Forest

Before applying any classification model to the dataset, existing solutions to the classification problem on the original ACDC challenge were investigated. Among the top-performing submissions, nearly all relied on neural network architectures to achieve high accuracy. However, one notable exception was the method described in Khened et al.[3], which achieved competitive results without relying on deep learning techniques. Instead, their approach was based on a Random Forest classifier, demonstrating that strong performance could still be achieved using a more traditional, interpretable model.

Random Forests offer a strong balance between performance and interpretability, which makes them particularly appealing for this task. Compared to deep learning models, which often function as black boxes, Random Forests provide useful insights through mechanisms like feature importance and decision path analysis, crucial aspects when working in medical domains. At the same time, they outperform simpler models like single decision trees by reducing overfitting and increasing robustness. As an ensemble method, Random Forests combine the predictions of multiple trees to capture non-linear relationships in the data while remaining stable and resistant to noise.

5.1 Data and Features

Before training the model, two preprocessing steps were applied to the data: shuffling and feature scaling. The training labels in the dataset were assigned in a sequential manner, grouped by class. To avoid introducing bias in the cross-validation process, the data was shuffled to ensure a more representative distribution of classes across folds.

Although Random Forests do not require feature scaling, since they rely on thresholds rather than distance metrics, standard scaling was still applied to all features. This decision was made to maintain compatibility with other machine learning models that might be explored in future work, particularly deep learning methods, which can be sensitive to feature magnitudes. Neural networks, for instance, often rely on gradient-based optimization and activation functions that perform best when input features are on a similar scale.

The final set of 13 features was selected based on domain knowledge and guidance from previous literature, which emphasized the diagnostic value of specific functional and anatomical metrics. These features include ventricular volumes at different cardiac phases, myocardial volume and thickness, as well as various derived ratios that have been used in clinical studies to characterize cardiac dysfunction. Although more features were calculated in earlier preprocessing steps, only those considered most clinically relevant and consistent with findings in the literature were retained for model training.

To better understand relationships between the selected features, a correlation matrix was computed (Figure 8). While some features, such as end-diastolic and end-systolic volumes, show strong correlations, they were deliberately retained when they provided complementary clinical insight. Even if the model may treat highly correlated variables similarly, clinicians often interpret them through different lenses. Preserving this level of interpretability helps ensure the model’s decisions remain aligned with clinical reasoning.

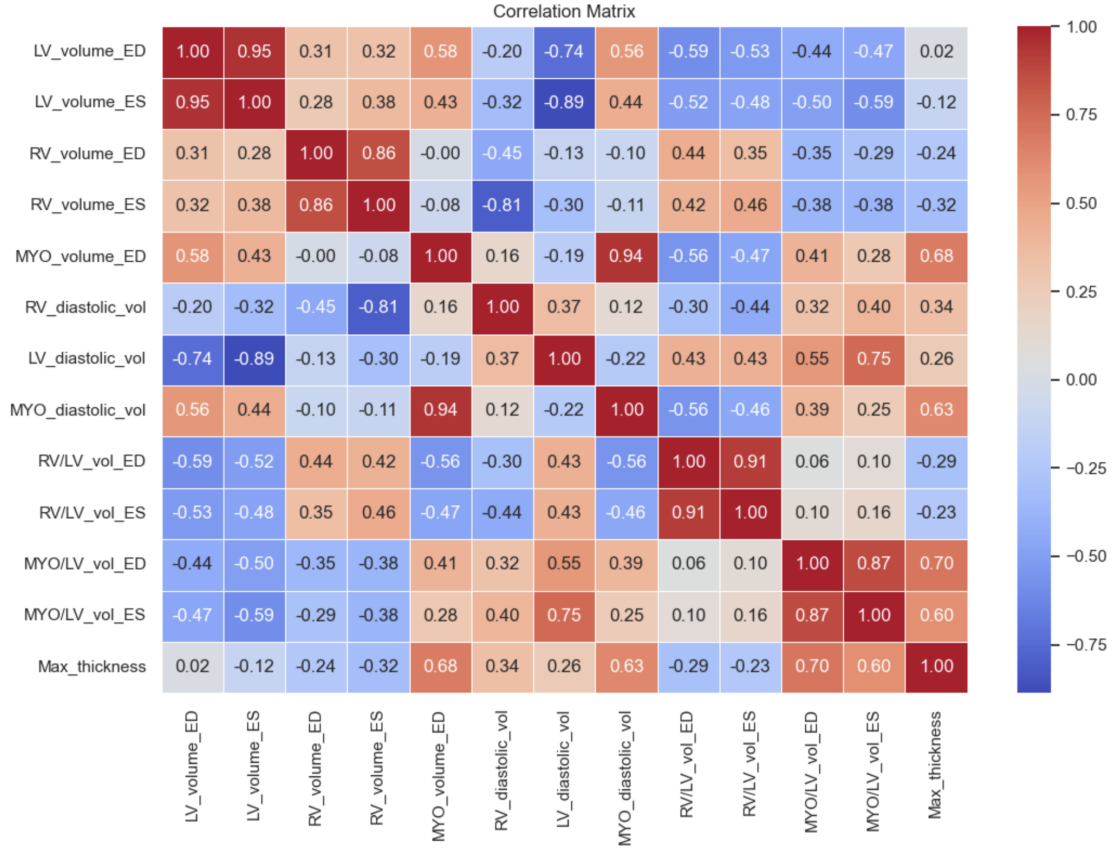


Figure 8: Correlation matrix of the selected features. Strong correlations exist among several pairs, especially between systolic and diastolic volumes of the same chamber, but all features were retained when clinically justified.

5.2 Modelling Method

With the preprocessed features selected and scaled, a Random Forest classifier was trained and optimized using a grid search over a range of hyperparameters. The tuning process was carried out using the entire training dataset, with a 5-fold cross-validation scheme to evaluate different parameter combinations and reduce the risk of overfitting.

The hyperparameters considered during tuning were as follows:

- `max_depth`: [2, 3, 4, 5, None]
- `min_samples_leaf`: [1, 2, 3]
- `min_samples_split`: [2, 3, 4]
- `max_features`: [2, 3, 4]
- `n_estimators`: [75, 100, 125, 150]

The selection of some of these intervals was guided by the work of Khened et al. [3], which reported strong classification results using a Random Forest classifier composed of 100 trees. Inspired by this, the number of estimators was centered around 100, and additional ranges were explored to fine-tune model complexity and feature usage.

For evaluation, multiple performance metrics were considered: accuracy, precision (weighted), recall (weighted), and F1-score (weighted). However, model selection during grid search was based on the accuracy score.

After completing the grid search, the following hyperparameter configuration was found to yield the best results:

Parameter	Value
max_depth	4
max_features	2
min_samples_leaf	1
min_samples_split	2
n_estimators	75

Once the best model was trained, feature importances were extracted to understand which variables most influenced the classifier’s decisions. The resulting importances are shown in Figure 9. As expected, several of the most influential features aligned with those highlighted in prior studies.

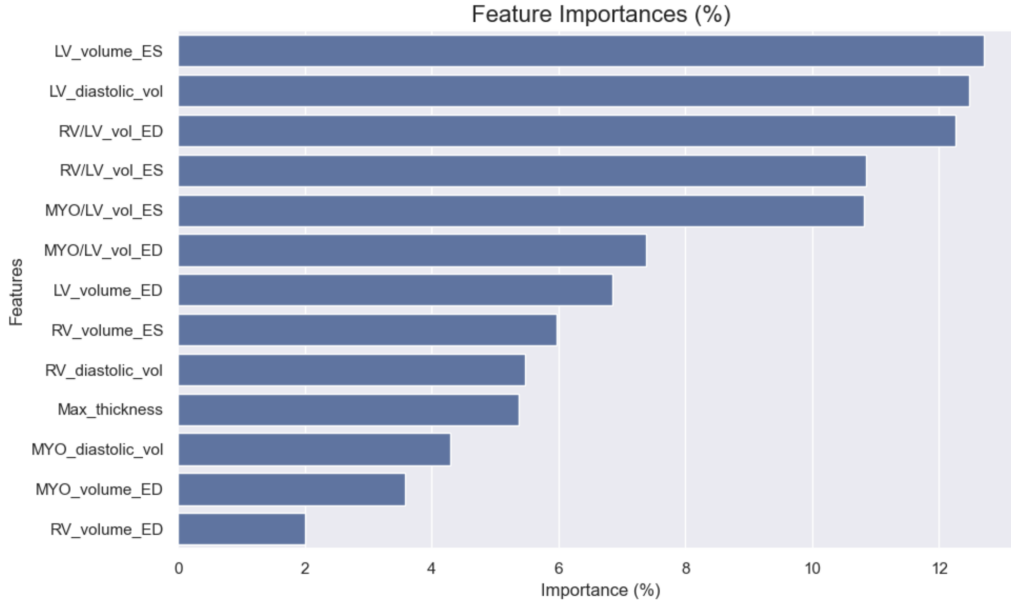


Figure 9: Feature importance of the trained Random Forest model. The most influential features correspond well with clinical expectations based on previous literature.

Despite some features being correlated, their importances reflect that they contribute in distinct ways to classification. This confirms the value of retaining them, especially given their individual diagnostic interpretations in clinical practice.

Finally, the best model (with the selected hyperparameters) was retrained on the entire training dataset and used to predict labels for the test set. These predictions formed the final submission to the Kaggle leaderboard.

6 Results

The performance of the Random Forest classifier was first evaluated using 5-fold cross-validation on the training set. The best model configuration, identified through grid search, achieved a weighted accuracy of 0.96 across the validation folds. This high score indicates that the model was able to effectively learn discriminative patterns in the data and generalize well within cross-validation splits.

To better understand how the model performed across different classes, a confusion matrix was generated (Figure 10). It shows that most classes were correctly classified, with only a few errors between specific categories.

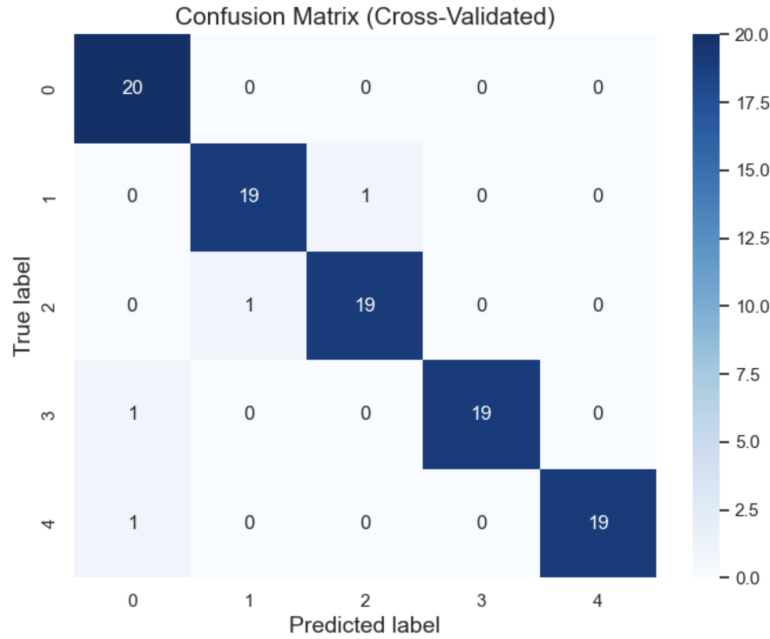


Figure 10: Confusion matrix on the training set after cross-validation. Most classes are accurately classified, with few misclassifications.

Beyond internal validation, the trained model was applied to the test data provided in the Kaggle challenge. The resulting predictions achieved an accuracy of 0.85 on the public leaderboard. Although this score is slightly lower than the one obtained during cross-validation, it still demonstrates strong generalization ability. The drop is likely due to differences in data variability or unseen patient cases in the test set.

7 Conclusion

This concludes the project for the IMA205 course, which applied machine learning methods to a real-world medical imaging classification problem using the ACDC dataset. Compared to the more guided practical labs, this project allowed for a more realistic application of the concepts covered in class. It involved going through the full machine learning pipeline, including data pre-processing, feature extraction, model selection, hyperparameter tuning, and evaluation.

One of the most challenging and instructive parts of the project was the extraction and selection of features from segmented medical images. Preparing these features in a way that they could be effectively used by traditional machine learning models required careful design and attention to clinical relevance. Nonetheless, the final model delivered strong results, showing that this approach was successful.

There are still several ways the model could be improved. Although the Random Forest gave good results, other ensemble methods like XGBoost or LightGBM might perform even better. It could also be useful to try more advanced feature engineering based on domain knowledge, or even combine traditional machine learning with deep learning techniques to see if that leads to better results.

Overall, this project not only reinforced key machine learning concepts but also highlighted the challenges and rewards of working with medical data in a more applied context.

References

- [1] Olivier Bernard, Alexandre Lalande, Clément Zotti, Florence Cervenansky, Xiayu Yang, Pheng-Ann Heng, and Pierre-Marc Jodoin. Deep learning techniques for automatic mri cardiac multi-structures segmentation and diagnosis: Is the problem solved? *IEEE Transactions on Medical Imaging*, 37(11):2514–2525, 2018.
- [2] Loic Le Folgoc and Pietro Gori. Ima205 challenge 2025. <https://kaggle.com/competitions/ima-205-challenge-2025>, 2025. Kaggle.
- [3] Mahendra Khened, Varghese Alex, and Ganapathy Krishnamurthi. Densely connected fully convolutional network for short-axis cardiac cine mr image segmentation and heart diagnosis using random forest. In Mihaela Pop, Maxime Sermesant, Pierre-Marc Jodoin, Alain Lalande, Xiaohai Zhuang, Guang Yang, Alistair Young, and Olivier Bernard, editors, *Statistical Atlases and Computational Models of the Heart. ACDC and MMWHS Challenges*, pages 140–151, Cham, 2018. Springer International Publishing.
- [4] Jelmer M. Wolterink, Tim Leiner, Max A. Viergever, and Ivana Išgum. Automatic segmentation and disease classification using cardiac cine mr images. In Mihaela Pop, Maxime Sermesant, Pierre-Marc Jodoin, Alain Lalande, Xiaohai Zhuang, Guang Yang, Alistair Young, and Olivier Bernard, editors, *Statistical Atlases and Computational Models of the Heart. ACDC and MMWHS Challenges*, volume 10663 of *Lecture Notes in Computer Science*, page automatic insert or look up actual page numbers, Quebec City, Canada, 2017. Springer.